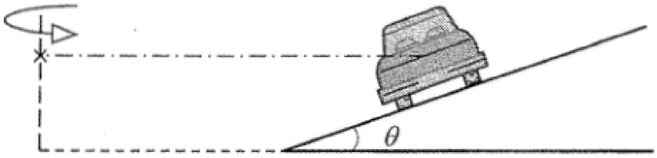
ch11 Uniform Circular Motion

2015 Q10

*10.



The figure shows the rear view of a car of mass m which travels along a circular road banked with an angle θ to the horizontal. The car moves at a certain speed such that it experiences **no frictional force along the inclined surface**. Which of the following represents the centripetal force on the car ?

A. $mg\sin\theta$

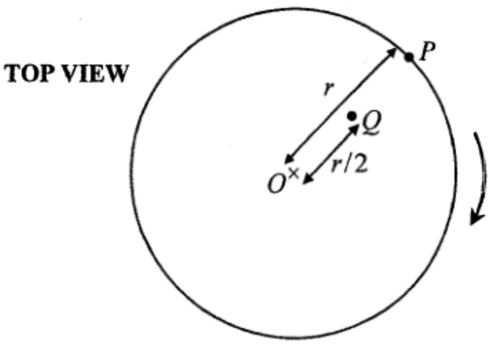
B. $mg\sin\theta\cos\theta$

C. $\dfrac{mg\cos\theta}{\sin\theta}$

D. $\dfrac{mg\sin\theta}{\cos\theta}$

2016 Q13

*13. Particles P and Q are fixed at distances r and $r/2$ respectively from the centre O of a horizontal circular platform which is rotating uniformly as shown.

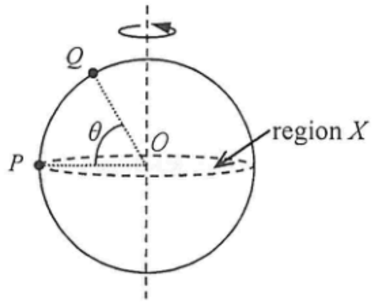


The ratio of the acceleration of P to that of Q is

A. 1 : 2.
B. 2 : 1.
C. 1 : 4.
D. 4 : 1.

2020 Q9

*9. Particles P and Q are fixed on the surface of a sphere rotating about a vertical axis passing through the centre O of the sphere as shown. The horizontal shaded region X divides the sphere into two halves. P is at the edge of region X while Q is at an angle of elevation θ above region X.



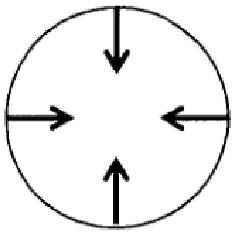
Find the ratio of the centripetal acceleration of P to that of Q.

A. $1 : \cos\theta$
B. $1 : \sin\theta$
C. $\cos\theta : 1$
D. $\sin\theta : 1$

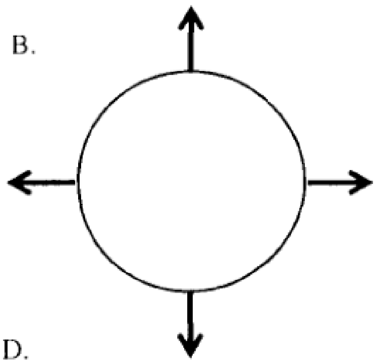
2021 Q9

*9. A particle is moving clockwise (top view) in a horizontal circle with uniform speed. Which top view diagram below correctly shows the net force acting on the particle at various positions ?

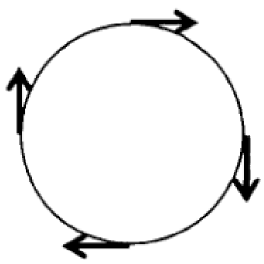
A.



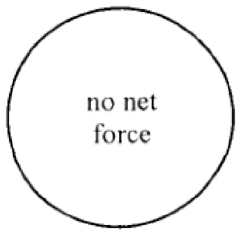
B.



C.

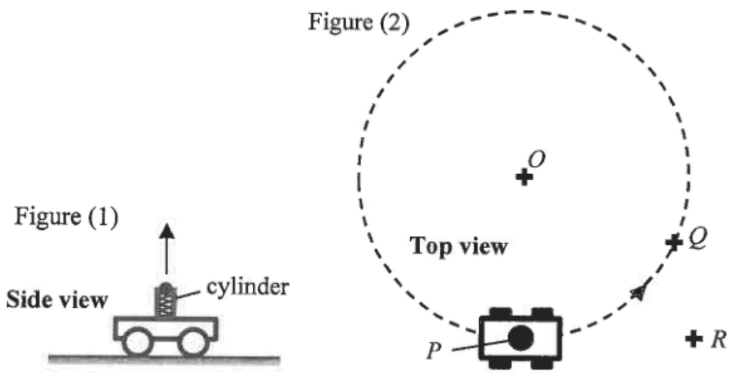


D.



2022 Q9

*9. A trolley with a spring-loaded launcher in Figure (1) can project a ball vertically upward. The trolley travels on the ground at a constant speed in a horizontal circle (with centre O) as shown in Figure (2).

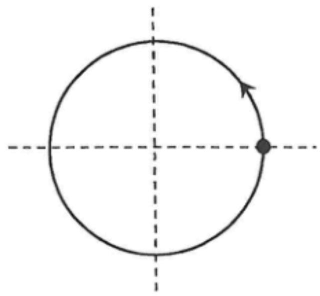


The ball is projected when the trolley is at point P. After some time, the ball falls back to the ground as the trolley just reaches point Q. Neglecting air resistance, where would the ball land ?

A. O
B. P
C. Q
D. R

2024 Q12

*12.

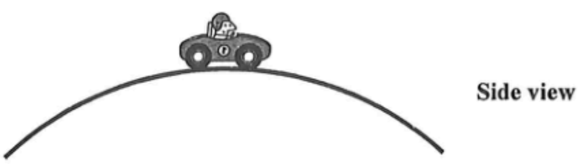


The figure shows a particle performing uniform circular motion. Which of the following correctly indicates the directions of its velocity and acceleration at the instant shown ?

	velocity	**acceleration**
A.	↑	←

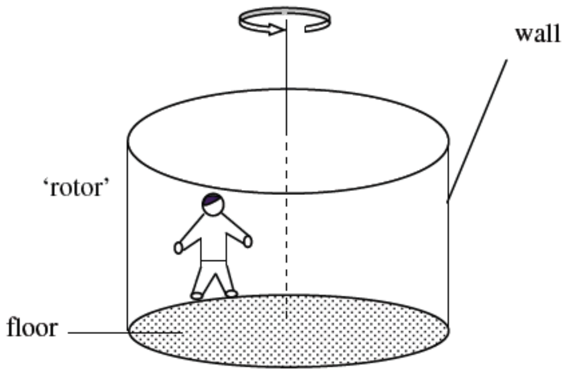
2025 Q11

*11. A car of weight W travels with constant speed on a circular humpback bridge. At the moment when the car passes the top of the bridge, the normal reaction acting on it by the bridge is R. What is the centripetal force acting on the car at that moment ?



A. $W + R$
B. $W - R$
C. W
D. R

*15.

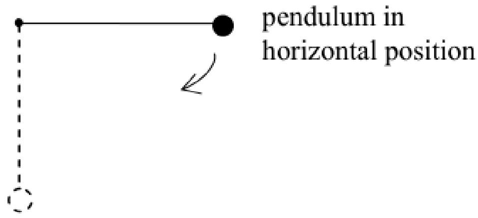


A man is rotating with constant speed inside a cylindrical 'rotor' and he remains pressed against the wall. The floor of the 'rotor' is smooth. Which of the following forces provides the centripetal force for the man ?

A. the weight of the man
B. the frictional force from the wall
C. the normal reaction from the wall
D. the supporting force from the floor

sap Q14

*14.



A simple pendulum is held at rest in a horizontal position. It is then released with the string taut. Which statement about the tension in the string is **not correct** when the pendulum reaches its vertical position ?

A. The tension equals the weight of the pendulum bob in magnitude.
B. The tension attains its greatest value.
C. The tension does not depend on the length of the pendulum.
D. The tension depends on the mass of the pendulum bob.